DocVerify — AI-Based Document Verification System

Phase 8 Scientific Remediation, Verification & Final Re-Evaluation Handover Report

Project Title	DocVerify: AI-Based Identity Document Forgery Detection
Project Root	`E:\DocverifyAi\`
Milestone	Phase 8 — Scientific Remediation & Final Frozen Re-Evaluation
Date of Compilation	October 18, 2026
Audit Basis	Phase 7A Independent Technical Audit (22 Findings Evaluated)
Remediation Status	**100% COMPLETE — NEW FROZEN BENCHMARK ESTABLISHED**

1. Executive Summary & Definitive Benchmark Comparison

In Phase 8, all 10 confirmed technical and statistical vulnerabilities uncovered during the Phase 7A Independent Technical Audit were systematically remediated. The previous 91.88% Phase 6 benchmark is permanently preserved as a historical pre-remediation baseline. The remediated Phase 8 multi-modal architecture was frozen and evaluated exactly once on the untouched 320-document test partition (151 Genuine, 169 Forged).

Milestone / Model	Threshold	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FN / FP
Phase 2 Global Baseline	0.50	79.69%	82.91%	77.51%	80.12%	0.9112	38 / 27
Phase 6 Benchmark (Historical)	0.50	*91.88%*	*95.54%*	*88.76%*	*92.02%*	*0.9780*	19 / 7
Phase 8 Balanced Policy	**0.72**	**91.25%**	**91.23%**	**92.31%**	**91.76%**	**0.9677**	**13 / 15**
Phase 8 Strict Anti-Fraud	**0.80**	**92.50%**	**98.01%**	**87.57%**	**92.50%**	**0.9677**	**21 / 3**
Phase 8 Conservative KYC	**0.50**	**90.31%**	**87.50%**	**95.27%**	**91.22%**	**0.9677**	**8 / 23**

2. Audit Issues Remediated (Root Causes & Concrete Fixes)

- **AUD-PATCH-01 (Bounding Box Parsing Bug):** Resolved silent dictionary parsing mismatch. Engineered src/bbox_parser.py with parse_sidtd_bbox() which parsed 100% of the 1,222 SIDTD forged bounding boxes with zero fallbacks.
- **AUD-PATCH-02 (Invalid EXP-06 Claim):** Officially invalidated the 99.38% test claim on static crops. Retrained models/phase8/dedicated_patch_model.pth on true sensor crops, achieving an honest **98.44% test accuracy**.
- **AUD-PATCH-03 (Checkpoint Mutation):** Replaced shallow .copy() with copy.deepcopy(), ensuring restored weights strictly match best validation epoch.
- **AUD-FUSE-01 (Feature Mismatch):** Extracted validation and test features using identical canonical layout priors; eliminated all Phase 4 oracle GT crops.
- **AUD-SHORT-02 (JPEG / ELA Shortcut):** Discovered that ELA separation collapses from Cohen's d=1.52 to d=0.07 under matched Q90 compression. Regularized fusion weights to prevent compression artifact domination.

• **AUD-TPL-01 (Template Filename Leak):** Engineered src/template_matcher.py using aspect ratio, native sensor scale, and OCR keywords (achieved 100.0% autonomous accuracy with zero filename hints).

• **AUD-CAL-01 (Truthful Calibration):** Fitted true Isotonic Regression on validation; Brier score improved from 0.0492 to 0.0370; test ECE held at 0.0484.

• **AUD-CAL-02 (Threshold Semantics):** Corrected operational hierarchy: Conservative (0.50) <= Balanced (0.72) <= Strict (0.80).

• **AUD-CAM-01 (Grad-CAM Distortion):** Implemented CorrectedGradCAM exact inverse scaling, reducing spatial offset error to 1px.

• **AUD-ENG-01/02 (App Stability):** Fixed Streamlit stale state on upload switch and added guaranteed unlinking of temporary report PNGs.

3. Statistical Rigor (Bootstrap CIs & McNemar Paired Test)

To establish scientific defensibility, non-parametric bootstrap resampling (5,000 iterations) and McNemar's paired test were executed on the frozen test partition (N=320):

Metric	Mean Estimate	95% Bootstrap Confidence Interval	Standard Error
Accuracy	91.22%	**[87.81%, 94.38%]**	0.0168
Precision	91.25%	**[86.83%, 95.40%]**	0.0221
Recall (Sensitivity)	92.24%	**[88.00%, 95.98%]**	0.0205
F1-Score	91.72%	**[88.46%, 94.67%]**	0.0159
ROC-AUC	96.75%	**[94.79%, 98.44%]**	0.0094

McNemar Paired Test: Tested against Phase 2 baseline on the same 320 test samples. Contingency: Both Correct=243, P2 Correct/P8 Wrong=12, P8 Correct/P2 Wrong=49, Both Wrong=16. Test statistic **Chi2 = 21.25**, **p-value = 4.04e-06 (< 0.0001)**. The performance gain of Phase 8 over the global baseline is statistically significant.

4. Automated Testing & Verification Suite

The newly introduced regression test suite (scripts/test_phase8.py) executed 14 comprehensive functional, statistical, and memory tests with **100% pass rate (14/14)**: CMYK conversion, autonomous template detection, UNKNOWN fallback, SIDTD bbox parsing, patch checkpoint deepcopy, feature schema ordering, calibration Brier/ECE, threshold monotonicity, Grad-CAM spatial geometry, OCR failure fallback, forensic consistency, PDF compilation, oversized memory guards, and stale session prevention.

5. Scientific Limitations & Claim Discipline

• **Decision-Support Scope:** DocVerify is an AI-assisted decision-support system for probabilistic forgery risk estimation. It does not legally establish identity or verify physical document possession.
• **Error Characteristics:** The system has an empirical false alarm rate of 9.93% (15/151) at the balanced threshold (0.72) driven by dense guilloche backgrounds, and a false negative rate of 7.69% (13/169) on single-character inpainting. At the Conservative KYC policy (threshold 0.50), missed forgeries drop to 4.73% (8/169).
• **Operational Deployment:** Human-in-the-loop inspection is recommended for all documents evaluated in the MEDIUM RISK category.